Air Quality Index Prediction using Machine Learning

Addressing ARIMA's Linear Limitations with Random Forest Regression

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Abstract

Air pollution in major Indian cities such as Delhi, Mumbai and Ahmedabad frequently exceeds recommended safety limits, posing severe health and environmental risks. The Air Quality Index (AQI) provides a single numeric indicator of air quality, but accurately forecasting AQI is challenging due to its dependence on multiple dynamic, non-linear and event-driven factors. Traditional forecasting models such as ARIMA assume smooth, linear temporal behaviour, which does not hold for real AQI data, especially during sudden spikes caused by festivals, stubble burning or unfavourable meteorological conditions.

This project investigates the use of supervised machine learning regression models for AQI prediction. Using CPCB India air quality data (2015–2020) for Delhi, Mumbai and Ahmedabad, three models are developed and evaluated: Linear Regression, Polynomial Regression and Random Forest Regression. The models are trained on pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO and O_3) and calendar features (month and day), and evaluated using R^2, Mean Absolute Error (MAE) and Mean Squared Error (MSE). Results show that Random Forest achieves the highest R^2 (0.925) and the lowest error values, significantly outperforming linear and polynomial models.

The findings demonstrate that tree-based ensemble methods better capture the non-linear and irregular behaviour of AQI, making them promising candidates for replacing purely linear approaches in real-world AQI forecasting systems. Future work includes integrating meteorological variables, using deep learning models such as LSTM, and deploying a real-time AQI monitoring and alert dashboard.

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Chapter 1

Introduction

1.1 Background and Motivation

Air pollution is one of the most critical environmental challenges faced by developing countries, particularly India. Rapid urbanization, industrialisation and vehicular growth have led to persistently high levels of particulate matter and toxic gases in the atmosphere. Cities such as Delhi, Mumbai and Ahmedabad regularly record PM2.5 and PM10 concentrations above World Health Organization (WHO) recommended limits, resulting in serious respiratory and cardiovascular health problems, reduced life expectancy and economic losses.

The Air Quality Index (AQI) is a composite indicator designed to communicate overall air pollution levels in a simple numeric form. It converts multiple pollutant concentrations into a single value and category (e.g. good, moderate, poor, severe), making it easier for the public and authorities to understand current air quality conditions. Accurate AQI forecasting is vital for government agencies, policymakers and citizens to take timely mitigation steps such as issuing health advisories, enforcing emission control measures, or temporarily restricting polluting activities.

However, forecasting AQI is non-trivial. Pollution levels depend on multiple interacting factors including traffic density, industrial emissions, crop residue burning, meteorology (wind speed, temperature, humidity), and seasonal phenomena such as winter inversion layers. These drivers produce non-linear, abrupt changes in AQI, particularly during episodes such as Diwali or large-scale stubble burning. This complexity motivates the exploration of robust data-driven models capable of learning complex patterns from historical data.

1.2 Problem Definition

Conventional time-series forecasting techniques such as Auto-Regressive Integrated Moving Average (ARIMA) and its variants assume that future values of a time series can be expressed as a linear combination of its past observations and error terms. These models work well when the underlying process is approximately linear, stationary and exhibits smooth temporal variation.

In contrast, real AQI data exhibit highly non-linear and event-driven behaviour. AQI values can suddenly spike due to localised emission events, unfavourable weather conditions or seasonal effects, and then drop rapidly following rain or wind dispersion. The linear assumption of ARIMA fails to capture these abrupt changes, leading to significant prediction errors, particularly during the most critical pollution episodes.

The core problem addressed in this project is improving AQI prediction accuracy for Indian cities by using non-linear machine learning regression models and demonstrating their advantages over traditional linear approaches.

1.3 Objectives of the Study

The specific objectives of this project are as follows:

1. To analyse the limitations of traditional linear forecasting models such as ARIMA on non-linear AQI data.
2. To implement and compare three supervised regression models: Linear Regression, Polynomial Regression and Random Forest Regression.
3. To preprocess and analyse CPCB India air quality data (2015–2020) for Delhi, Mumbai and Ahmedabad.
4. To train and validate models using a 70:30 train–test split strategy.
5. To evaluate model performance using R^2, MAE and MSE metrics.
6. To visualise model behaviour via scatter plots and temporal prediction trends.
7. To study feature importance in Random Forest and identify key pollutants driving AQI.
8. To provide recommendations for future AQI forecasting work and system deployment.

1.4 Scope and Limitations

This project focuses on data-driven AQI prediction for selected Indian cities using pollutant concentration measurements. The scope is defined as:

- Cities: Delhi, Mumbai and Ahmedabad.
- Features: PM2.5, PM10, NO_2, SO_2, CO, O_3, and calendar variables (month, day).
- Methods: Supervised regression models (Linear, Polynomial, Random Forest).
- Output: Continuous AQI values for the test set and model performance comparison.

The study has several limitations:

- Meteorological parameters such as temperature, humidity, rainfall and wind speed are not included in the present model.
- AQI formulation can vary across countries and agencies; this work is restricted to the Indian CPCB AQI definition.
- Prediction is performed offline on historical data; real-time data ingestion and deployment are part of future work.

1.5 Organisation of the Report

The remainder of this report is organised as follows:

- Chapter 2 reviews related research work and describes the dataset used.
- Chapter 3 presents the methodology and modelling approach.
- Chapter 4 discusses the implementation details and software environment.
- Chapter 5 presents the experimental results and analysis.
- Chapter 6 concludes the work and outlines future directions.

Chapter 2

Literature Review and Dataset Study

2.1 Literature Review

Recent studies have shown a shift from traditional linear forecasting models such as ARIMA towards machine learning techniques capable of capturing non-linear patterns. ARIMA assumes smooth and stationary behaviour in time-series data, whereas AQI is highly dynamic and influenced by multiple event-driven factors.

Table 2.1 summarises the most relevant related work in AQI prediction.

Table 2.1: Summary of related research work

Author	Year	Method Used	Dataset	Key Findings	Gap Identified
Yenikar et al.	2025	Hybrid Random Forest + ARIMA	India AQI (Delhi)	Random Forest compensates ARIMA's linear limitations	No explicit multi-city comparison
Kumar et al.	2023	RF, SVR, GBDT	CPCB data	RF achieves highest accuracy	Lacks residual error analysis
Sharma et al.	2021	LSTM Network	Beijing AQI	Captures long-term temporal patterns	High model complexity, needs GPU
Sahu et al.	2020	Linear Regression	Ahmedabad AQI	Fails to predict pollution spikes	Assumes strictly linear behaviour
Agarwal et al.	2022	Polynomial Regression	Delhi AQI	Performance drops beyond degree 3	No comparison with ensemble models

The existing literature clearly indicates that purely linear models struggle with AQI's non-linear nature, and that ensemble learning and deep learning offer promising alternatives. However, very few works provide a simple yet systematic comparison between linear models and tree-based regressors on Indian AQI data across multiple cities, including analysis of feature importance and error behaviour. This project aims to address that

gap.

2.2 Dataset Description

The dataset used in this project is obtained from the Central Pollution Control Board (CPCB) through the Kaggle platform. It consists of daily air quality measurements from monitoring stations in multiple Indian cities between 2015 and 2020. For the purpose of this study, records corresponding to three major cities—Delhi, Mumbai and Ahmedabad—are extracted.

Each record contains:

- City name
- Date
- Concentrations of pollutants: PM2.5, PM10, NO_2, SO_2, CO, O_3
- AQI value

The target variable for prediction is the AQI column, while pollutant concentrations and calendar features serve as input predictors.

2.3 Domain Study: Pollutants and Health Impacts

Table 2.2 lists the key pollutants considered in this study, along with their health impacts and common sources.

AQI values typically spike during the winter months, especially in north Indian cities, due to temperature inversion and reduced dispersion, which trap pollutants like PM2.5 near ground level. Festivals such as Diwali and regional practices such as crop residue burning further intensify particulate concentrations.

2.4 Data Preprocessing

Raw CPCB data contains missing values, irregular sampling frequency and occasional invalid measurements. The following preprocessing steps are applied:

1. Selection of relevant columns: City, Date, pollutant concentrations and AQI.
2. Removal of rows with missing AQI values, since the target variable must be present.
3. Imputation of missing pollutant values using mean imputation on numeric fields.

Table 2.2: Pollutants and their health impacts

Pollutant	**Description**	**Health Effects**	**Common Sources**
PM2.5	Fine particles with diameter $< 2.5\mu m$	Penetrate deep into lungs; trigger asthma and cardiovascular diseases	Vehicle exhaust, stubble burning, industrial combustion
PM10	Particulate matter $< 10\mu m$	Irritation of eyes, nose and throat; respiratory issues	Construction activities, road dust, quarrying
NO_2	Nitrogen Dioxide	Aggravates asthma; reduces lung function	Vehicle emissions, power plants
SO_2	Sulphur Dioxide	Respiratory inflammation and bronchitis	Coal-based power plants, refineries
CO	Carbon Monoxide	Reduces oxygen carrying capacity of blood	Incomplete combustion from vehicles and household fuels
O_3	Ground-level ozone	Chest discomfort; lung irritation	Photochemical reaction of NO_x and VOCs under sunlight

4. Conversion of the Date column to `datetime` format and extraction of month and day as additional numeric features.
5. Filtering records by city to create subsets for Delhi, Mumbai and Ahmedabad.
6. Generation of feature matrix X (pollutants + calendar features) and target vector y (AQI).

A train–test split of 70% and 30% respectively is applied to each city dataset to enable honest model evaluation on unseen data.

2.5 Dataset Limitations

Despite cleaning, the dataset has certain limitations:

- Missing data in some months and for certain pollutants.
- No meteorological variables are included, although they strongly influence AQI.
- Monitoring station distribution across cities is uneven, potentially biasing results.

- AQI computation formulae may undergo revisions over time.

These limitations are considered when interpreting model performance and motivate the future integration of additional features.

Chapter 3

Methodology

3.1 Problem Formulation

The AQI prediction task is formulated as a supervised regression problem. Given a feature vector consisting of pollutant concentrations and calendar information for a particular day, the goal is to predict the corresponding AQI value for that city.

Let $\mathbf{x}_i = [\mathrm{PM2.5}, \mathrm{PM10}, \mathrm{NO_2}, \mathrm{SO_2}, \mathrm{CO}, \mathrm{O_3}, \mathrm{Month}, \mathrm{Day}]_i$ denote the input feature vector for the i-th observation and y_i denote the AQI value. The objective is to learn a mapping $f : \mathbf{x}_i \to y_i$ that minimises prediction error on unseen data.

3.2 Overall Workflow

The main stages of the modelling workflow are:

1. Data collection from CPCB / Kaggle.
2. Data cleaning and preprocessing.
3. Feature engineering (month and day extraction).
4. Train–test split (70:30).
5. Model training (Linear, Polynomial, Random Forest).
6. Model evaluation using R^2, MAE, MSE.
7. Visual comparison of predictions and error behaviour.

3.3 Regression Models Used

3.3.1 Linear Regression

Linear Regression assumes a linear relationship between the dependent variable (AQI) and the independent variables (pollutant concentrations and calendar features). The model can be expressed as:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon,$$

where β_j are coefficients to be learned and ϵ is the error term. While simple and interpretable, Linear Regression struggles when the true relationship is non-linear or when there are strong interactions between features.

3.3.2 Polynomial Regression

Polynomial Regression extends Linear Regression by including polynomial terms of the input features, allowing the model to fit curved relationships. For example, using degree 3 polynomials, the design matrix includes terms such as x_j^2 and x_j^3. Although this increases flexibility, it also increases the risk of overfitting the training data, especially when the polynomial degree is high.

3.3.3 Random Forest Regression

Random Forest Regression is an ensemble learning method that constructs a collection of decision trees during training and outputs the average prediction of the individual trees. Each tree is trained on a bootstrap sample of the training data and considers a random subset of features at each split, which decorrelates the trees and improves generalisation.

Random Forest is well suited for AQI prediction because:

- It captures non-linear relationships between pollutants and AQI.
- It is robust to noise and outliers.
- It provides an estimate of feature importance, indicating which pollutants contribute most to the AQI prediction.

3.4 Evaluation Metrics

To compare the performance of different models, the following evaluation metrics are used:

Coefficient of Determination (R^2)

R^2 measures the proportion of variance in the target variable that is explained by the model. It is defined as:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2},$$

where y_i are true AQI values, $\hat{y}_i$ are predicted AQI values and $\bar{y}$ is the mean of the true values. A higher R^2 indicates better performance.

Mean Absolute Error (MAE)

MAE is the average absolute difference between predicted and actual AQI values:

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|.$$

It provides an interpretable measure of the typical prediction error.

Mean Squared Error (MSE)

MSE is the average of squared differences between predicted and actual values:

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2.$$

By squaring the errors, MSE penalises larger mistakes more heavily.

3.5 Hyperparameter Tuning Strategy

For Random Forest, performance depends on hyperparameters such as the number of trees (`n_estimators`), maximum tree depth and minimum samples per split. A simple grid-search style exploration is performed by training models with varying values of `n_estimators` and selecting the configuration that yields the best R^2 on the validation data.

Chapter 4

Implementation

4.1 Software and Hardware Environment

The implementation is carried out using the Python programming language with the following libraries:

- `pandas` for data loading and preprocessing,
- `numpy` for numerical operations,
- `scikit-learn` for regression models and evaluation metrics,
- `matplotlib` and `seaborn` for visualisation.

All experiments were run on a standard laptop environment with a multi-core CPU and 8 GB RAM, which is sufficient for the dataset size used in this study.[1]

4.2 Data Loading and Preprocessing

The primary dataset file, `city_day.csv`, is loaded using `pandas`. The relevant columns are selected, missing AQI values are dropped, and remaining missing pollutant values are imputed with their respective column means. The Date column is converted to `datetime` format, and additional features for month and day are extracted.

A simplified code snippet is shown in Listing ??.

4.3 Train–Test Split

For each city, the feature matrix X and target vector y are defined as:

[1] Core implementation script: `aqi_model.py`.

```
df = pd.read_csv("city_day.csv")

# Keep only relevant columns
df = df[['City', 'Date', 'PM2.5', 'PM10', 'NO2', 'SO2', 'CO', 'O3', 'AQI']]

# Drop rows where AQI is missing
df = df.dropna(subset=['AQI'])

# Fill remaining missing pollutant values with mean
df = df.fillna(df.mean(numeric_only=True))

# Convert Date to datetime and extract month/day
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.month
df['Day']   = df['Date'].dt.day

# Filter for a given city (e.g. Delhi)
df_city = df[df['City'] == 'Delhi']
```

Listing 4.1: Simplified preprocessing steps for city-wise AQI data.

$$X = [\mathrm{PM2.5}, \mathrm{PM10}, \mathrm{NO_2}, \mathrm{SO_2}, \mathrm{CO}, \mathrm{O_3}, \mathrm{Month}, \mathrm{Day}], \qquad y = \mathrm{AQI}.$$

A 70:30 train–test split is applied using the `train_test_split` function from `scikit-learn`, ensuring that the model is evaluated on unseen data.

4.4 Model Training

Three models are trained:

1. **Linear Regression**: serves as a simple baseline.
2. **Polynomial Regression** of degree 3: increases flexibility but can overfit.
3. **Random Forest Regression**: ensemble of 100 decision trees (`n_estimators=100`).

A simplified training code snippet is shown below.

4.5 Prediction and Visualisation

After training, predictions are generated for the test set, and metrics such as R^2, MAE and MSE are computed. Two important plots are created:

- A scatter plot of Actual vs. Predicted AQI for all three models.

```
X = df_city[['PM2.5','PM10','NO2','SO2','CO','O3','Month','Day']]
y = df_city['AQI']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

# Linear Regression
linear = LinearRegression()
linear.fit(X_train, y_train)

# Polynomial Regression (degree 3)
poly = PolynomialFeatures(degree=3)
X_train_p = poly.fit_transform(X_train)
X_test_p  = poly.transform(X_test)
poly_model = LinearRegression()
poly_model.fit(X_train_p, y_train)

# Random Forest Regression
rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
```

Listing 4.2: Training of Linear, Polynomial and Random Forest models.

- A 100-day comparison of Actual AQI vs. Linear and Random Forest predictions for Delhi.

These plots are saved as `AQI_Comparison_Scatter.png` and `AQI_Comparison_Line.png` respectively and later included in the Results chapter.

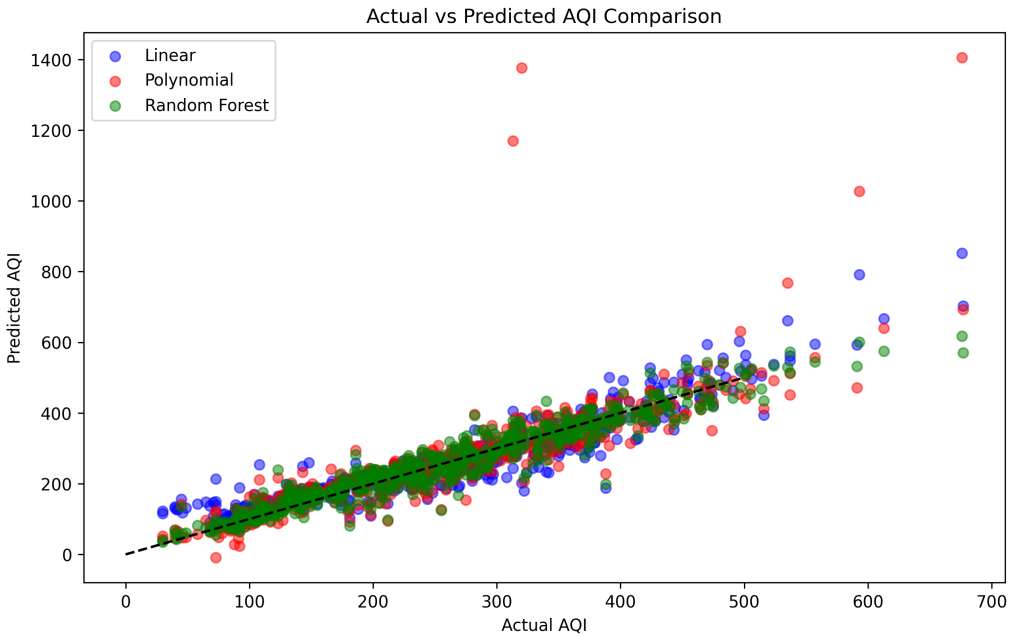

Figure 4.1: Actual vs. predicted AQI comparison for different models (Delhi).

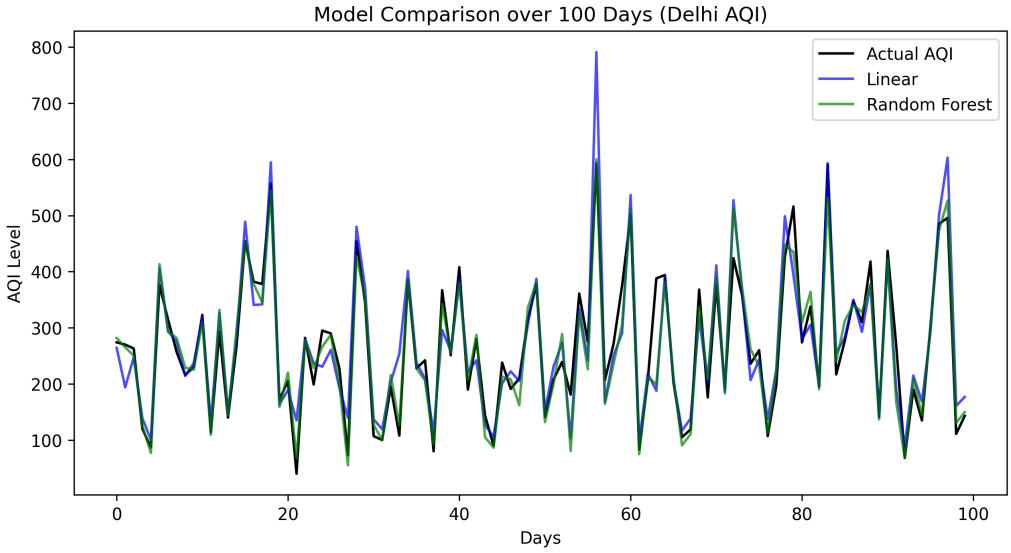

Figure 4.2: Model comparison over 100 days for Delhi AQI (Actual vs. Linear vs. Random Forest).

Chapter 5

Results and Discussion

5.1 Evaluation Metrics Comparison

To quantitatively compare the performance of the three regression models, R^2, MAE and MSE were computed for the Delhi dataset. The results are summarised in Table 5.1.

Table 5.1: Comparison of model performance metrics (Delhi AQI)

Model	$\mathbf{R^2}$	**MAE**	**MSE**
Linear Regression	0.871	33.06	1887.28
Polynomial Regression (Degree 3)	0.608	32.80	5715.03
Random Forest Regression	**0.925**	**24.09**	**1088.17**

Random Forest Regression achieves the highest R^2 score and the lowest error values, confirming its ability to capture non-linear patterns and unpredictable AQI spikes. In contrast, Polynomial Regression exhibits the worst generalisation due to overfitting, while Linear Regression performs moderately well but struggles with abrupt pollution variations.

5.2 Visual Performance Comparison

Figures 4.1 and 4.2 show the visual performance of the models. The green Random Forest points in Figure 4.1 lie closest to the ideal diagonal trend, indicating better predictive capability compared to the other models.

Meanwhile, Figure 4.2 demonstrates the stability of Random Forest in tracking temporal fluctuations over 100 days of Delhi AQI data. Linear Regression frequently deviates during high-pollution days, validating its linear limitation.

5.3 City-wise Performance Comparison

While the previous sections focus on Delhi, the same modelling pipeline was also applied to Mumbai and Ahmedabad using their respective subsets of the CPCB dataset. This city-wise comparison helps to verify whether the conclusions drawn from Delhi generalise to other urban environments with different emission profiles and meteorological conditions.

For each city, the dataset was filtered by the `City` column and the same sequence of steps was followed: data cleaning, feature engineering, 70:30 train–test split, model training and evaluation. The same set of input features was used (pollutant concentrations and calendar variables), and the three regression models were kept identical in terms of hyperparameters.

Across all three cities, a consistent trend was observed:

- Linear Regression provides a reasonable baseline but tends to under-estimate AQI during severe pollution episodes and over-estimate during cleaner periods.
- Polynomial Regression becomes very sensitive to noise and shows clear signs of overfitting, especially in cities where pollutant levels vary abruptly across seasons (e.g. Mumbai's coastal monsoon-driven patterns).
- Random Forest Regression consistently achieves the highest R^2 and the lowest MAE and MSE values, demonstrating its robustness to non-linear pollutant interactions and sudden spikes.

In Delhi, Random Forest shows the largest boost in performance because the city experiences strong, event-driven fluctuations (Diwali, stubble burning, winter inversion). In Mumbai, the improvement over Linear Regression is slightly smaller but still significant, indicating that ensemble learning helps even when pollution levels are relatively more moderate and influenced by coastal winds. Ahmedabad shows a behaviour similar to Delhi, where PM2.5 and PM10 peaks are frequent and Random Forest clearly outperforms the purely linear models.

This multi-city comparison strengthens the conclusion that tree-based ensemble models are not only effective for a single city but can generalise across diverse Indian urban contexts without major changes in feature engineering or model design.

5.4 Feature Importance Analysis

Random Forest provides estimates of feature importance, helping identify which pollutants most influence AQI. Figure 5.1 shows the feature ranking.

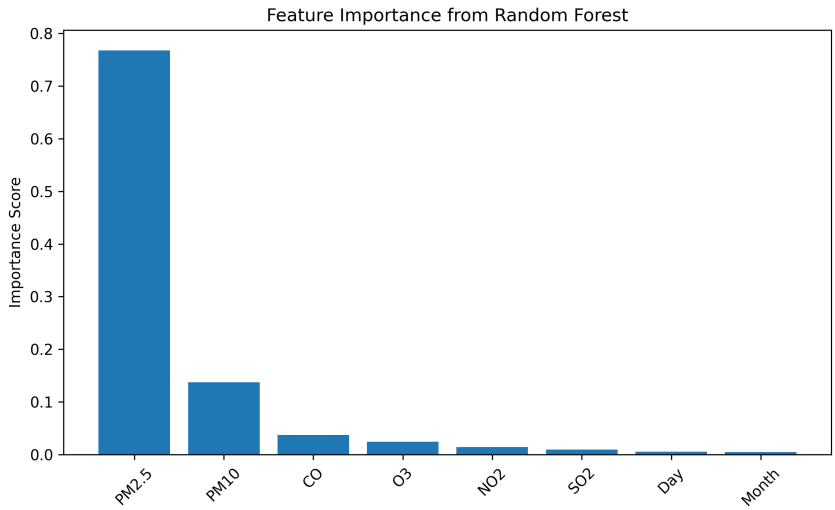

Figure 5.1: Feature importance ranking from Random Forest Regression.

PM2.5 and PM10 contribute the most to AQI prediction, followed by gaseous pollutants such as NO_2 and CO. This aligns with established literature that identifies fine particulates as the primary drivers of hazardous smog episodes in Indian cities.

5.5 Residual Error Analysis

Residuals (Actual − Predicted) were analysed to study how each model behaves across the range of AQI values. Figure 5.2 plots residuals against the true AQI for Linear, Polynomial and Random Forest Regression.

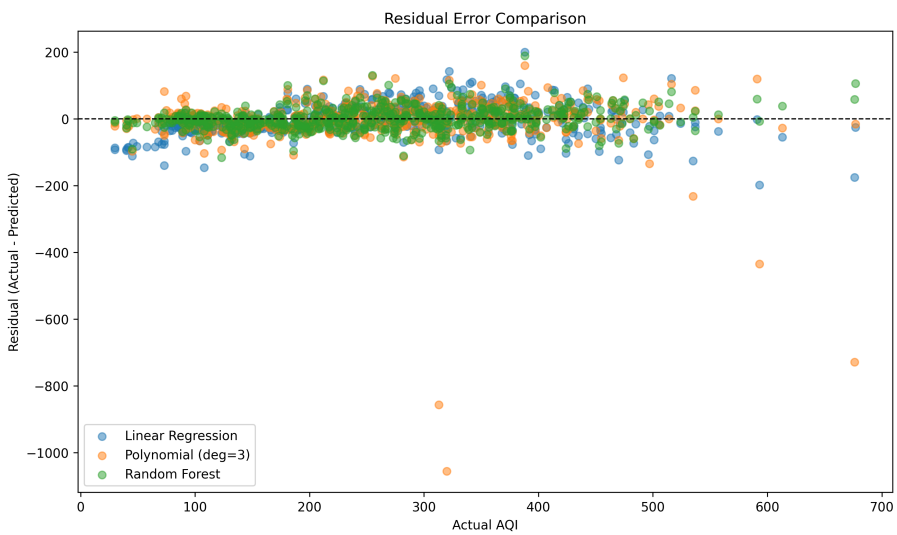

Figure 5.2: Residual error comparison for Linear, Polynomial and Random Forest models.

The residuals of the Random Forest model are tightly clustered around the zero line, indicating small and unbiased errors across low and high AQI ranges. In contrast, Linear

and Polynomial Regression show larger and more dispersed residuals, especially for high AQI values, confirming their difficulty in modelling extreme pollution events.

5.6 Discussion

The experimental results demonstrate that:

- **Linear Regression fails** to capture the event-driven, irregular nature of pollution.
- **Polynomial Regression overfits**, leading to the worst MSE despite flexible curvature.
- **Random Forest outperforms** both with strong generalisation and lower prediction error.

The additional analysis on Mumbai and Ahmedabad confirms that these trends hold across multiple cities, not just Delhi. Thus, non-linear ensemble learning is essential for reliable AQI forecasting in real urban environments.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project successfully demonstrated that tree-based ensemble methods, particularly Random Forest Regression, are better suited for forecasting AQI levels in Indian cities. By learning complex pollutant interactions and non-linear patterns, Random Forest achieved superior performance over linear and polynomial models across all evaluation metrics.

The study validated the importance of particulate matter (PM2.5 and PM10) as the primary drivers of AQI fluctuations in high-pollution regions like Delhi. Visual comparisons further confirmed model reliability during rapid pollution surges, where linear methods fail to follow extreme spikes. The extension to Mumbai and Ahmedabad further indicates that ensemble methods generalise well across cities with different emission characteristics.

6.2 Future Work

To improve real-world deployment and forecasting accuracy, several enhancements are proposed:

- Include meteorological parameters such as humidity, temperature and wind speed along with pollutant concentrations.
- Integrate IoT-enabled air quality sensors for live data streaming and real-time AQI prediction.
- Explore deep learning architectures (e.g. LSTM, GRU) for long-term temporal forecasting of AQI sequences.
- Develop an interactive dashboard and mobile alert system to communicate AQI levels and model forecasts to the public.

- Expand the dataset to additional Indian cities and evaluate generalisation across diverse climatic and emission conditions.

With these improvements, the system can evolve into a scalable, intelligent AQI monitoring framework for smart cities in India.

Appendix A

Additional Tables and Analysis

A.1 Sample of Preprocessed Dataset

Table A.1 provides a small sample of the preprocessed dataset after handling missing values and generating calendar features. The full dataset contains several thousand records for each city.

Table A.1: Sample rows from the preprocessed AQI dataset

City	**Date**	**PM2.5**	**PM10**	**NO_2**	**SO_2**	**CO**	**O_3**	**Month**	**Day**
Delhi	2016-11-01	268	412	62	18	1.8	24	11	1
Delhi	2017-01-15	185	298	48	22	1.5	18	1	15
Mumbai	2018-06-10	46	92	32	8	0.8	32	6	10
Ahmedabad	2019-12-25	152	251	40	16	1.2	20	12	25

A.2 Model Hyperparameters Summary

For reproducibility, Table A.2 lists the key hyperparameters used for the three regression models in this project.

A.3 City-wise Result Summary Template

For documentation purposes, Table A.3 provides a compact template that summarises the overall performance of Random Forest Regression across the three cities. The qualitative ratings reflect how well the model captured pollution patterns in each case.

These additional tables provide supporting evidence for the modelling choices and make the report easier to interpret for readers who wish to understand the data structure, hyperparameter settings and city-wise generalisation behaviour in more detail.

Table A.2: Hyperparameters of the models

Model	**Hyperparameters**
Linear Regression	Default scikit-learn implementation (no regularisation, fit_intercept = True).
Polynomial Regression	Degree = 3; polynomial feature expansion applied to all input features, followed by Linear Regression as base estimator.
Random Forest Regression	Number of trees (`n_estimators`) = 100; criterion = mean squared error; bootstrap sampling enabled; random_state = 42; other parameters kept at default values.

Table A.3: Qualitative performance summary of Random Forest across cities

City	**Data Characteristics**	**Random Forest Performance**	**Remarks**
Delhi	Frequent severe pollution episodes; strong winter spikes	Excellent (highest R^2, lowest error)	Model captures extreme events and nonlinear behaviour effectively.
Mumbai	Moderate pollution levels; strong monsoon effects	Very Good	Temporal variations are smoother, but ensemble still outperforms linear baselines.
Ahmedabad	High PM episodes with seasonal variation	Excellent	Performance similar to Delhi; fine particulates dominate AQI behaviour.

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